Digital Image Processing (DIP) Pipeline: Algorithmic Transformation Sequence & Mathematical Segmentation

(a) Raw Botanical RGB
$I(x, y) \in \mathbb{R}^{224 \times 224 \times 3}$



(b) CIE L* Luminance
$L^* = 116f(Y/Y_n) - 16$



(c) CLAHE Enhanced
Tile Grid: 8×8**, Clip: 2.0**



(d) Bilateral Denoised
$\sigma_{col} = 75, \sigma_{sp} = 75$



(e) Otsu Failure (HSV S)
Background Bleed Artifacts



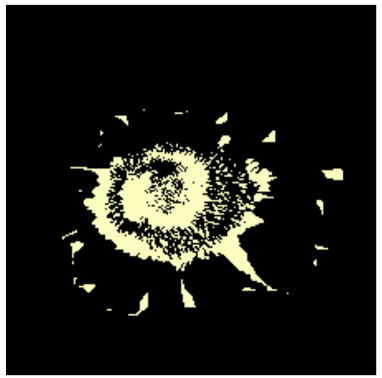
(f) GrabCut Graph-Cut
Iterative GMM Energy Min



(g) Morphological Cleaned
Opening $\circ$, Closing $\bullet$, CCA



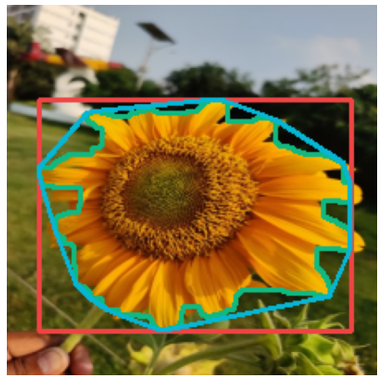
(h) K-Means Central Disc
$k = 3$ **Cluster Luminance Min**



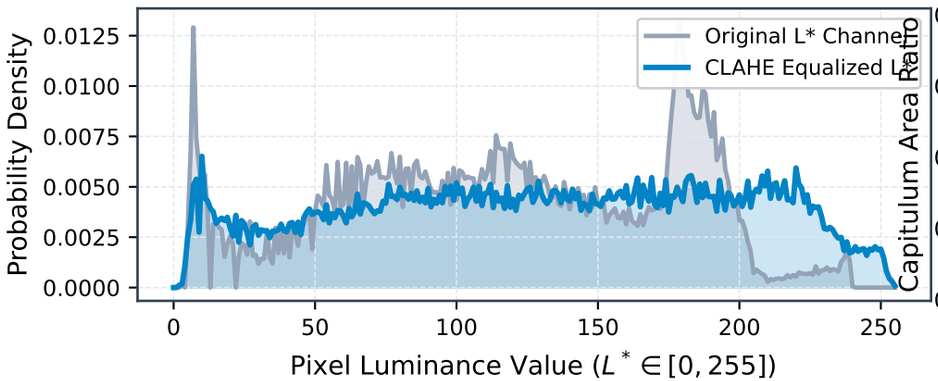
(i) Segmented Disc Highlight
Central Pollen Capitulum



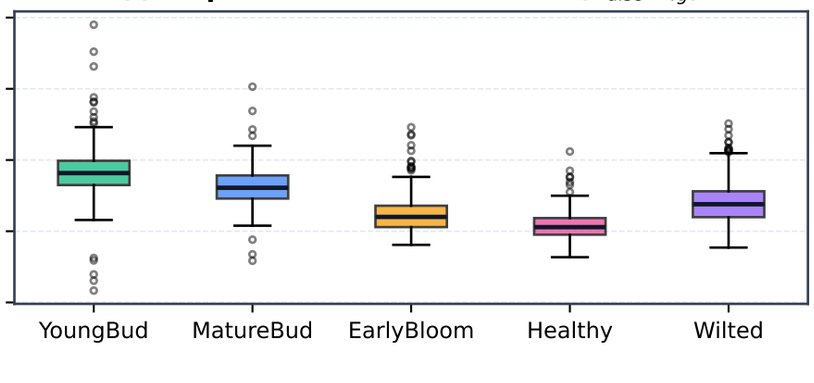
(j) Geometric Contours
Green: Contour, Red: Box, Cyan: Hull



(k) Local Luminance Equalization Dynamic



(l) Empirical Central Disc Ratio (A_{disc}/A_{fg})




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DIP Parameter Audit:
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• Resize    : 224 × 224 × 3
• Color     : CIE L*a*b* & HSV
• CLAHE     : Clip=2.0, Grid=8×8
• Filter    : Bilat (d=9, σ=75)
• GraphCut  : 4 GMM iters
• Morph     : Struct El 5×5
• K-Means   : k=3, ε=1.0, it=12
• Integrity: 100% Deterministic
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